POLITECNICO MILANO 1863

BASI DI DATI 1 – Anno Accademico 2024-2025

Proff. Anna BERNASCONI, Stefano CERI, Sara COMAI

Call n. 5 - 12/09/2025, **DURATION: 2 hours**

Exercise 1 – SQL (13pt)

The following basic data schema (keys underlined) describes the policies of an insurance agency:

CUSTOMERS(TaxID, Name, Address, City)
INSURANCEPOLICIES(CustomerTaxID, LicensePlate, Premium)
PAYMENTS(CustomerTaxID, LicensePlate, Date, PaymentMethod)
VEHICLES(LicensePlate, OwnerTaxID, Province, EngineSize, Year)
MOTORCYCLES(LicensePlate, OwnerTaxID, Province, EngineSize, Year)

Formulate the following queries in SQL:

A. Write the SQL command to create the PAYMENTS table, making appropriate reasonable assumptions about domains and constraints. (1pt)

B. Return customers who only own motorcycles. (4pt)

C. Identify the customers spending the most on policies for vehicles they do not own. (4pt)

D. Find customers who have exactly one car policy and one motorcycle policy in their name. (4pt)

Exercise 2 – Formal languages (6pt)

Express query B in optimized Relational Algebra (3 pt) and Datalog (3 pt).

Exercise 3 – Theory question (1pt):

Explain the effect of the "on update cascade" clause in the definition of a foreign key constraint in SQL.

Exercise 4 – Conceptual and logical design (13pt)

- *The project evaluation is based on its completeness, correctness, legibility, minimality and clarity. Remember to specify the **identifiers** of all the entities and the **cardinality** of all the associations.*
- *Arrange the conceptual project and the logical project on the inside pages of the sheet; the logical project is also a graph, with nodes for tables and arcs for referential integrity, which must reflect the position of entities and relations in the conceptual project.*

An educational portal offers a wide range of courses with high-quality content. To participate in the courses, users must register on the platform, providing their personal data, an email address, and a password, and then they can enroll in individual courses. The courses are organized by subject area (e.g., engineering, computer science, design, soft skills) and always have a title.

Each course has been recorded by one or more teachers, whose first name, last name, and a brief description are stored, and can be offered in English, Italian, or both languages. Each course has a start and end date and can be repeated several times during the year, but a user can only take one of its editions. The courses are divided into modules, each with a sequential number, a title, and a duration. Each module includes a set of video lessons and a self-assessment quiz. Some courses require other courses as prerequisites to facilitate understanding of the content.

The quizzes consist of a predetermined number of questions, each with a set of answers, only one of which is correct. The system keeps track of the modules taken by a user and, for each self-assessment quiz, keeps track of how many correct answers have been given so that it can provide a final certificate.